

Functional Requirements Specification (FRS)

Project: Online Hotel Booking System (OHBS)

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Purpose: Requirements foundation for manual testing, STLC activities, and Selenium automation

1. Business Overview

Business Problem

Our hotel chain currently relies on third-party booking platforms, resulting in:

- High commission costs
- Limited control over customer experience
- Inconsistent booking data

Business Objective

Develop an Online Hotel Booking System that allows customers to:

- Search hotels
- View availability and pricing
- Make and manage reservations directly

This system will reduce third-party dependency and improve customer engagement. **In-**

Scope

- Hotel search and booking
- User registration and login
- Booking management (view, cancel)
- Payment processing (basic) **Out-of-Scope**
- Loyalty points
- Mobile application
- Third-party travel integrations

2. User Personas

Guest User (Unregistered)

- **Goals:** Search hotels, view prices, explore availability
- **Pain Points:** Cannot book without registration

Registered Customer

- **Goals:** Book rooms, manage bookings, receive confirmation
- **Pain Points:** Slow booking process, unclear cancellation policies **Hotel**

Admin (Future Phase)

- **Goals:** View bookings (read-only for current scope)

3. Functional Requirements

User Registration

Description:

The system shall allow new users to create an account.

Inputs:

- Full Name
- Email Address
- Password
- Confirm Password **Business Rules:**
- Email must be unique
- Password must contain:
 - Minimum 8 characters
 - At least one number and one special character

User Login

Description:

The system shall allow registered users to log in using email and password.

Exceptions:

- Invalid credentials
- Locked account after 5 failed attempts

Hotel Search

Description:

The system shall allow users to search hotels based on:

- Location
- Check-in Date
- Check-out Date
- Number of guests

Room Selection

Description:

The system shall display available rooms with:

- Room type
- Price per night
- Amenities
- Availability status

Booking Confirmation

Description:

The system shall allow a logged-in user to book a selected room.

Post-Conditions:

- Booking confirmation ID generated
- Email confirmation sent to registered email

Payment Processing

Description:

The system shall allow users to make payment using:

- Credit Card
- Debit Card
- Payment must be successful to confirm booking

View Booking

Description:

The system shall allow users to view booking details in “My Bookings”.

Cancel Booking

Description:

The system shall allow users to cancel a booking before check-in date.

Business Rule:

- Cancellation not allowed on check-in date or after

4. Non-Functional Requirements

Performance

- Page load time shall not exceed 3 seconds under normal load **Security**
- Passwords must be encrypted
- HTTPS protocol must be used **Usability**
- System shall be usable on major browsers (Chrome, Edge, Firefox)
- Error messages must be user-friendly

Reliability

- System availability: 99% uptime

5. Acceptance Criteria

Login – Valid Credentials

Given the user is on the login page

When the user enters valid email and password

Then the system shall redirect the user to the dashboard

Login – Invalid Credentials

Given the user is on the login page

When the user enters invalid credentials

Then an error message "Invalid email or password" shall be displayed

Hotel Search

Given the user is on the home page

When the user enters valid search criteria
Then the system shall display a list of available hotels

Booking Confirmation

Given the user has selected a room
And payment is successful
When the booking is confirmed
Then a booking ID shall be generated And
confirmation email shall be sent

Cancel Booking

Given the user has an active booking
When the user cancels before check-in date
Then the booking status shall be updated to "Cancelled"

6. Assumptions & Constraints

Assumptions

- Users have valid email ID
- Payment gateway sandbox used for training

Constraints

- Web-based application only
- English language only

7. Approval & Next Steps

Stakeholder Review

- Trainer acts as Business Analyst
- Feedback incorporated before final sign-off

Subsequent Usage

This document will be used to:

- Derive manual test cases
- Write Selenium automation scripts
- Perform functional and regression testing